

Organic Chemistry — Quick Notes (Test Document)

Page 1 of 3

Introduction to functional groups. Alkanes are saturated hydrocarbons with single carbon-carbon bonds. They follow the general formula C_nH_{2n+2} . Common reactions include free-radical halogenation under UV light, where a hydrogen is replaced by a halogen. Alkenes contain at least one carbon-carbon double bond and are more reactive than alkanes. Electrophilic addition is characteristic: for example, bromine water decolorizes when added to an alkene due to addition across the double bond. Markovnikov's rule predicts regioselectivity when adding HX to unsymmetrical alkenes. This document is synthetic content for backend PDF extraction and chunking tests. It spans multiple pages so pagination and text flow can be validated.

Organic Chemistry — Quick Notes (Test Document)

Page 2 of 3

Carbonyl chemistry overview. Aldehydes and ketones feature a polar C=O bond, making the carbon electrophilic. Nucleophiles such as cyanide, Grignard reagents, and hydride donors add to the carbonyl carbon.

Carboxylic acids and derivatives include esters, amides, and acid chlorides. Relative reactivity toward nucleophilic acyl substitution generally follows: acid chloride > anhydride > ester ~ acid > amide. Hydrolysis of esters under basic conditions is saponification.

Buffers in biological systems often involve weak acids and conjugate bases. The Henderson-Hasselbalch equation relates pH, pKa, and the ratio of conjugate base to acid. Titration curves show distinct equivalence points for polyprotic species.

Organic Chemistry — Quick Notes (Test Document)

Page 3 of 3

Laboratory safety reminders for instructional labs. Always wear appropriate PPE: splash goggles, lab coat, and closed-toe shoes. Know locations of eyewash, safety shower, and fire extinguisher. Never taste or smell chemicals directly; use wafting if odor must be assessed. Waste segregation matters: halogenated organics, aqueous metals, and non-halogenated solvents often go into different streams. Label all unknowns clearly. Document observations in a bound notebook with dates and initials.

These paragraphs are filler for multi-page PDF testing. Topics mentioned are educational summaries only and not a substitute for a real course syllabus.